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(54) Title: PROCESSES FOR CONVERSION OF WASTE MATERIALS

(57) Abstract: The technology concerns a unique and superior methodology for converting waste materials of various compositions into solid objects of superior properties.

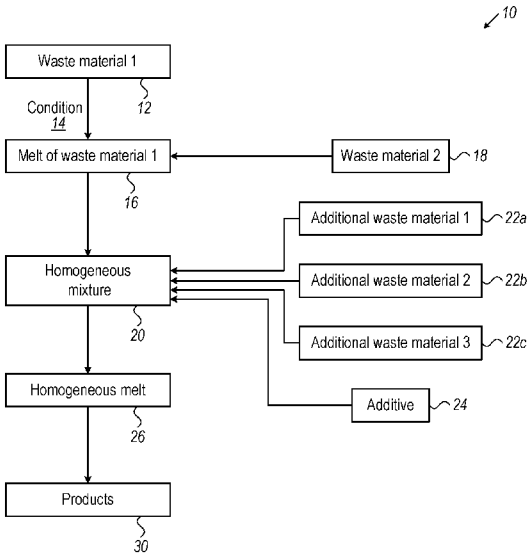


FIG. 1

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PROCESSES FOR CONVERSION OF WASTE MATERIALS

TECHNOLOGICAL FIELD

The invention generally contemplates objects and materials, such as construction materials, derived from waste materials and processes for manufacturing same.

BACKGROUND OF THE INVENTION

The conversion of waste materials into new products has been subject of research for many years. While material reusing typically aims at recovering some, if not all, of the material's original attributes, reacquiring a material's original properties may not be possible in certain cases and the material must therefore be transformed into a different reusable form or disposed of.

Organic waste material such as fruit and vegetable peelings, grass clippings, wood chippings and many food components are typically recycled by converting the organic waste into compost or liquid fertilizer and fuels. The use of such materials for producing hard objects or surfaces has seen little success.

BACKGROUND PUBLICATIONS

[1] US Patent Publication No. 2008/245269;

[2] US Patent Publication No. 2013/233206;

[3] US Patent Publication No. 2013/022823;

[4] US Patent Publication No. 2013/295394;

[5] US Patent Publication No. 2013/160674.

GENERAL DESCRIPTION

The inventors of the technology disclosed herein offer a unique and superior methodology for utilizing organic waste materials or certain decomposable materials as binders or filler materials in processes of manufacturing solid objects and materials that can be used for a variety of purposes. The inventors have developed a novel recyclizing and waste conversion process involving conversion of compositions of both municipal and industrial waste, i.e., post-consumer and post-industry wastes, which may comprise organic waste materials and food components, as well as polymer materials and other

waste components into aesthetic objects and functional objects and materials such as construction materials and construction objects as bricks, panels and rooftiles. A unique feature of the technology resides in the fact that while metallic components are not used, materials manufactured from such compositions have demonstrated mechanical properties that are surprising to those of cement-based materials.

In its broadest aspect, the invention concerns a composition for use in a process of manufacturing a 3D object or material, such as a construction material, wherein the composition comprises unprocessed waste free of metals.

The invention further provides a process for converting a waste composition, or a plurality of such compositions, into solid products, without requiring pretreatment of any sort.

As used herein, a ***composition*** used in manufacturing objects and materials according to the invention is a collection of waste materials that comprises any type of waste, excluding metals and dangerous materials (such as toxic material, explosives, etc). The term 'composition' is not intended to limit the collection of waste materials by any way or indicate any structuring or composing method. Rather, the term is used to encompass a mixture of waste components that are classified based on their origin, composition, type and other parameters, all being derived from discarded or disposed waste or such waste components intended for landfill disposal. A waste composition may thus include, for example, organic waste including food products, organic and inorganic oils such as cooking oils and industrial oils, salts (e.g., metal salts such as calcium carbonate, but not metallic materials), plastics and polymers, as well as other paper products, textiles, tires and other components depending on the origin of the waste. The amount or relative amount of each component in a composition used according to the invention may vary and may be tailored, e.g., by adding one or more component, in order to achieve a final object, material, or generally a product with a predetermined mechanical, chemical and visual attributes.

However, when employing a process of the invention, any waste composition may be utilized to produce solid products of superior properties as disclosed herein. Typically, the waste material need not be sorted nor cleaned or washed prior to use and may be used 'as is'. The composition may be in a form of a waste material that contains a mixture of different materials, each having a different profile, namely different mechanical, physical and/or chemical properties; or in a form of different waste compositions, each containing

a different type of materials. The different waste compositions may be obtained by sorting a general municipal or industrial waste into different components, i.e., plastics, glass, paper-based materials, organic waste etc, or may be provided as different types of materials from different industries, e.g., plastic production, paper production, organic waste, tire industry, etc. The different waste components may be used as separate components or admixed prior to processing.

A "***waste material***" is any such used, scrap, and/or discarded material, or any such material that directly originates from waste, or which is attributable to waste, or which is or comprises a composition that is directly and/or indirectly derived from recycled materials. The waste material may be any of the waste material components or groups of materials disclosed herein.

The waste material may be provided in any form, shape and size. For ease of handling, the waste material may be mechanically treated to reduce the size of the waste objects or material to small processable fragments or chunks or pieces, wherein optionally the fragments are of a size ranging between 1 to 10 mm.

The ***organic waste*** component present in industrial and municipal waste, and used in accordance with the invention, is typically waste foods and food components, as well as any organic material, as known in the art that is discarded in landfills. The waste foods and food products may be in a form of processed or natural food products and food components, in solid, gel or liquid forms.

In some embodiments, the waste is or comprises discarded food products.

In some embodiments, the organic waste comprises organic waste derived from processing of various food products, mainly fruits and vegetables. The organic waste may include waste of fruits and vegetables, e.g., olive waste such as orujo, yeast, and food additives such as thickening agents, emulsifiers, stabilizers, and others.

In some embodiments, the waste is or comprises a liquid, such as an edible oil. Non-limiting examples of edible oils include coconut oil, corn oil, canola oil, cottonseed oil, olive oil, palm oil, peanut oil, rapeseed oil, safflower oil, sesame oil, soybean oil, sunflower oil, almond oil, beech nut oil, Brazil nut oil, cashew oil, hazelnut oil, macadamia oil, mongongo nut oil, pecan oil, pine nut oil, pistachio oil, walnut oil, pumpkin seed oil, and others.

In some embodiments, the oil is an inedible oil. In some embodiments, the inedible oil is derived from a variety of vegetable oils.

Non-limiting examples of inedible oils include copaiba, jatropha oil, jojoba oil, milk bush, nahor oil, paradise oil, petroleum nut oil, pongamia oil, dammar oil, linseed oil, poppyseed oil, stillingia oil, tung oil, vernonia oil, karanja, mahua, rubber seed, cottonseed, neem, putranjiba, tobacco seed, polanga seed, cardoon, deccan hemp, castor, jojoba, moringa, cuphea, poon and others.

The organic waste may further comprise industrial byproducts or materials which may be solids or liquids. Industrial oils may include gear oils, air compressor oils, hydraulic oils, rock drill oils, turbine oils, engine oils, lubricants, and others.

Thus, the ***oil*** present in compositions of the invention is any oil or a wax material, being in a liquid or semiliquid form, and selected from edible, inedible and industrial oils, as defined herein. In some embodiments, the oil is selected from coconut oil, corn oil, canola oil, cottonseed oil, olive oil, palm oil, peanut oil, rapeseed oil, safflower oil, sesame oil, soybean oil, sunflower oil, almond oil, beech nut oil, Brazil nut oil, cashew oil, hazelnut oil, macadamia oil, mongongo nut oil, pecan oil, pine nut oil, pistachio oil, walnut oil, pumpkin seed oil, copaiba, jatropha oil, jojoba oil, milk bush, nahor oil, paradise oil, petroleum nut oil, pongamia oil, dammar oil, linseed oil, poppyseed oil, stillingia oil, tung oil, vernonia oil, gear oils, air compressor oils, hydraulic oils, rock drill oils, turbine oils, engine oils, lubricants, and others.

In some embodiments, the waste is or comprises a powder material, which may be an edible solid or food component.

In some embodiments, the powder material is clay, calcium carbonate, talc, metal salts or metal complexes (neither being or comprising a metal atom in zero valency), dextrin, maltodextrin, starch, textured vegetable proteins, casein, proteins, flour, corn flour, wood flour, wood chips, and others.

The powder material is not ash.

Plastic waste and generally polymeric materials may be any such material that is a product of polymerization reactions or curing reactions which are used for manufacturing objects, elements or features of any shape and size, elements or apparatuses. Such plastic and polymeric waste materials may include a variety and an unlimited collection of polymeric materials, including thermoplastics, thermosets and elastomers. These may be selected from natural polymeric, semisynthetic polymers or synthetic polymers used in clothing, sporting objects, electronic and photonic technologies, packaging materials and containers, insulation of water systems or

electrical and thermal systems, construction and structural elements, painted objects and adhesives, lubricants and oils, waxes, vehicles, household items, objects used in the medical or cosmetic arenas, protective gear, industrial elements and machinery, and others.

The plastic or polymeric waste may be or may comprise rubber of any industrial form, low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polyurethane, acrylonitrile butadiene styrene, cellulose acetate, cellulose acetate butyrate, ethylene-vinyl acetate, fluorinated ethylene propylene, fluoropolymers, high-performance plastics, Pearloid, perfluoroalkoxy alkane, perfluoroether, Plastisol, poly(ethyl methacrylate), poly(methyl methacrylate), polyphenylene sulfide, polyacrylonitrile, polyamide, polyamide-imide, polyaryletherketone, polybutadiene, polybutylene succinate, polybutylene terephthalate, polycaprolactone, polycarbonate, polychlorotrifluoroethylene, polycyclohexylene -dimethylene terephthalate, polyester, polyether ether ketone, polyetherimide, polyetherketoneketone, polyhydroxyalkanoates, polyhydroxybutyrate, polyimide, polyketone, polylactic acid, polymethylpentene, polyoxymethylene, polyphthalamide, polystyrene, polysulfone, polyvinyl ester, polyvinyl fluoride, polyvinylcarbazole, bakelite, epoxy and related polymers, furan resins, melamine resins, Novolak, polybenzoxazine, polyhexahydrotriazine, polyisocyanurate, silicone, urea-formaldehyde, vinyl ester resin and others.

Tires and recycled tires may, but not necessarily, form a part of the products formed according to the invention. The tires are typically made of an elastomer, which forms the tread of the tire. Recycled tires or shredded tires used according to the invention are free of the metallic components which support the tire structure. Thus, tires used according to the invention may be composed of various composites of rubber e.g., styrene-butadiene copolymer.

The waste may also contain building material waste derived from construction sites, demolition sites, or generally from the construction industry. Such waste may contain non-metallic composite panels, artificial stone, asphalt, bricks and building stones, concrete, composite materials free of metals, glass materials, insulation materials, polymeric materials, sand, soil-based materials, terracotta, wood and others.

The waste may further comprise glass and ceramic materials which may be derived from construction materials, glass industry, municipal waste, optic systems, glassware, etc. The glass may be laminated glass or non-laminated glass, insulated glass, low-e glass, fire resistant glass, self-cleaning glass, acoustic glass, low iron or low metal glass, silicate glass, and any other glass of any thickness, type, color, structure and composition.

Also, may be present in the waste used according to processes of the invention are metal-free electronic units or objects or elements, which in the absence of metal may comprise glass fibers, glass components, polymeric materials, and other organic materials such as adhesives, pigments, and others.

Paper waste in a form of a variety of paper products may also be part of the waste materials. Such paper products include anything that is cellulose based, or which is derived from pulp and/or may include synthetic or semi synthetic paper products. The waste products may include books, newspapers, magazines, posters, wrapping materials, pamphlets, maps, signs, labels, advertisements, notebooks, writing pads, envelopes, packaging materials, stationery, parchment, toilet paper, paper towels, paper plates and paper cups, beverage cartons, tea bags, food packaging, coffee filters, wallpaper and others.

Fabrics and textiles may also be present in compositions of the invention. The fabric or textile may be formed of a natural fiber, such as cotton, or from a semisynthetic or synthetic polymeric materials. The fabric or textile may be of any type used in the fabric or textile industry, for example in clothing, furnishing and upholstery, bedding, vehicle furnishings and objects, medical and cosmetic textile, and others.

As waste materials may comprise a variety of unknown components, compositions of waste used according to processes of the invention may additionally comprise adhesive materials, flame retardant materials, emulsifiers, stabilizers, antioxidants, waxes and lubricants, pigments and coloring agents, binders, biodegradable additives, blowing agents, filler of a variety of forms and amounts, light stabilizers, carbon black, and others.

Thus, generally speaking, products formed according to a process of the invention may be formed of post-consumer and post-industry waste compositions, which comprise any one or more of:

1- Polymeric products,

2- Glass and/or ceramic products;

3- Metal-free electronic products;

4- Building or construction products;

5- Paper products;

6- Organic products; and

7- Other unidentified components or any other material that may be found in a post-consumer or post-industry waste material, excluding a metal.

In some cases, products formed according to processes of the invention may thus comprise one or more of:

1- Polymeric waste materials,

2- Glass-based waste materials;

3- Metal-free electronic-based waste materials;

4- Construction-based waste materials;

5- Paper-based waste materials;

6- Organic waste materials; and

7- Any other unidentified components or any other material that may be found in a post-consumer or post-industry waste material, excluding a metal.

Each of the waste materials may be used untreated or 'as is' or may be pretreated:

-to remove metallic materials,

-to remove toxic materials,

-to reduce the product into a manageable or processable size by mechanical means such as shredding, cutting, crushing, pressing, etc., which do not impose chemical modification to the material, and/or

-to combine materials into a mixture of materials.

Compositions of the invention may further contain at least one additive, e.g., an organic additive, which may be selected amongst materials present in the waste or generally in food products or in naturally derived products. Where the additive is present, it may be part of the organic waste material or may be separately added in order to endow certain mechanical attributes to the manufactured product. In some embodiments, the at least one additive is added to a waste composition, under conditions as disclosed herein, even if an amount thereof may already be present in the waste material.

The additive, e.g., organic additive, may be selected from yeast, carbohydrates and natural gums. The additive may be added in a powder, solution, or suspension form in an amount between 0.5 and 2wt% relative to the total weight of the composition used.

The organic additive may be used to convert the waste mass into a solid continuous form, by potentially, but not necessarily, forming chemical associations between the various components. The chemical associations, if present, may be covalent, van der Walls, hydrogen bonding or any other association. While the composition is not homogenous in composition, size, chemical and mechanical attributes, and while the organic additive constitutes only between about 0.5 and 2 wt% of the total weight of the composition, the organic additive is capable of amalgamating or rendering continuous all components into a substantially monolithic mass. Presence of small amounts of the organic additive vastly improves the stability and mechanical properties of an object formed therefrom and thus the suitability of the manufactured objects for use under various conditions (indoors, outdoors, exposure to natural elements, etc).

In some embodiments, the additive is yeast, which may be present in the organic waste material or which may be added to the composition as a powder or as a suspension in water. The amount of yeast added may be between 0.5 and 2 wt%, measured based on the total weight of the composition. The yeast may be of any type known. Typically, strains of yeast commonly used in baking are of the species *Saccharomyces cerevisiae*. Alternatively, *Saccharomyces exiguus* (*S. minor*) is present or used, alone or in combination.

In some embodiments, the organic additive may be a carbohydrate which may be part of the organic waste or may be added separately, as a powder, a liquid or a solution/suspension in water, typically in an amount between 0.5 and 2wt% relative to the total weight of the composition. The carbohydrate may be any such that is present or used in food products, fruits and vegetables, including for example sugars such as glucose, galactose, fructose, xylose, sucrose, lactose, maltose, isomaltulose, trehalose, sorbitol, mannitol; oligosaccharides such as maltodextrins, raffinose, stachyose, fructo-oligosaccharides and others; polysaccharides such as starch, amylose, amylopectin, modified starches, glycogen, cellulose, hemicellulose, pectin, hydrocolloids and others.

In some embodiments, the organic additive may be selected from maltodextrin, dextrose, maltitol, erythritol, sorbitol, mannitol, modified starch (such as acid-treated starch, alkaline-treated starch, bleached starch, oxidized starch, enzyme-treated starch,

mono-starch phosphate, di-starch phosphate, phosphated di-starch phosphate, acetylated di-starch phosphate, starch acetate, acetylated di-starch adipate, hydroxypropyl starch, hydroxypropyl di-starch phosphate, hydroxypropyl di-starch glycerol, starch sodium octenyl succinate, acetylated oxidized starch, cationic starches, hydroxyethyl starch, carboxymethylated starches and others.

In some embodiments, the organic additive may be at least one or more natural gum, which may be provided as part of the organic waste or may be separately added, as a powder or a solution/suspension, in an amount of between 0.5 and 2wt%. The natural gum may be any one or more of alginic acid, beta-glucan, caranna, chicle, dammar gum, galactomannan, gellan gum, glucomannan, guar gum, gum anima, gum Arabic, gum guaicum, gum karaya, konjac, locust bean gum, mastic, myrrh, neem gum, psyllium, tragacanth, welan gum, xanthan gum and others.

The organic additive may be additionally or alternatively selected from clay, talc, dextrin, maltodextrin, starch, textured vegetable proteins, casein, proteins, flour, corn flour and others.

In some embodiments, the organic additive may be a combination of any of the aforementioned organic additives. Where a combination of additives is used, the amount of the additive combination may be between 0.5 and 2 wt%, or each of the additive may be provided in an amount between 0.5 and 2 wt%, relative to the total weight of the composition.

The organic additive is not ash.

Thus, compositions used according to the invention may comprise one or more of polymeric waste materials, glass waste materials, paper-based waste materials, organic waste materials (e.g., derived from food materials, oils and others as disclosed herein), and at least one additive, each of which as defined herein.

Thus, compositions used according to the invention generally comprise

-Sorted or unsorted waste materials, as disclosed herein; and

-one or more auxiliary materials selected amongst organic additives or other materials which may be present in the waste, as disclosed herein.

In some cases, waste compositions of the invention may comprise any one of:

a. colored plastic (e.g., in a form of chips or granules);
b. plastics and polymers (e.g., typically recycled);
c. black plastic;

d. metal-free cement (e.g., typically processed or recycled);
e. fabrics or textiles (e.g., in shredded or cut forms);
f. paper products (e.g., in shredded or cut forms);
tires (e.g., in shredded or cut forms); and
g. others.

Compositions used according to the invention may comprise any combination as exemplified in any one of **Tables 1** to **6**:

No.	Material
1	Colored plastic
2	Plastics and polymers
3	black plastic
4	metal-free cement
5	fabrics or textiles
6	paper products
7	tires
8	Metalloids

Table 1: Exemplary combination of waste materials

No.	Material
1	Plastic or polymeric material
2	Calcium carbonate
3	Oil
4	binder, as selected
5	Colored plastic
6	metal-free cement
7	tires
8	Metalloids

Table 2: Exemplary composition

No.	Material
1	Plastic or polymeric material

2	Calcium carbonate
3	Oil
4	binder, as selected
5	black plastic
6	metal-free cement
7	tires
8	Metalloids
9	Fabric or textile

Table 3: Exemplary composition

No.	Material
1	Plastic or polymeric material
2	Calcium carbonate
3	Oil
4	binder, as selected
5	Colored plastic
6	metal-free cement
7	tires
8	Paper products
9	Fabric or textile

Table 4: Exemplary composition

No.	Material
1	Plastic or polymeric material
2	Calcium carbonate
3	Oil
4	binder, as selected
5	Colored plastic
6	metal-free cement
7	tires
8	Metalloids
9	Fabric or textile

10	Paper products

Table 5: Exemplary composition

#	Binder	Construction	Paper waste	Oil waste	Fabric waste	Tires waste	Glass waste	"Black" plastic	Waste	Plastic waste	Sawdust	Gaff
1	X	X	X	X	X	X	X	X	X	X	X	X
2	X	X	X	X		X		X	X	X		
3	X	X	X	X				X	X	X	X	
4	X	X		X	X	X	X	X	X	X		
5	X	X	X	X		X	X		X	X		
6	X	X	X	X	X			X	X	X		
7	X	X		X		X	X	X	X	X		
8	X	X		X	X	X	X			X		
9	X	X	X	X				X		X		
10	X	X	X	X	X	X		X	X	X		X
11	X	X	X	X	X	X		X	X	X		
12	X	X	X	X	X				X	X	X	

Table 6: Additional exemplary compositions

Waste compositions, as disclosed herein, may be used in processes for producing composite materials for a variety of uses. As waste compositions contain waste materials which exhibit different stabilities or sensitivity profiles, mainly under high temperature conditions, processes of the invention have been tailored to allow for proper and effective incorporation of a variety of waste materials to produce stable solid objects or composite materials. Generally speaking, a process of the invention comprises a plurality of processing stages, each step designed to introduce a component of a waste material under conditions which permit its stable incorporation in a final melt or product.

In most general terms, a process of the invention utilizes waste materials that have not been pretreated, but for being reduced in size, e.g., by mechanical shredding or crushing, without substantially causing chemical decomposition or without introducing chemical modifications to the waste materials, to afford a more easily processable mass. Thus, a process of the invention comprises treating a selection of waste materials or waste material components under different conditions enabling their homogeneous distribution in a flowable melt to afford a solid object or a raw solid material, which may be used for manufacturing of construction units, such as building blocks, tiles and flat surfaces; or

ornamental objects. The various waste materials, while described as different classes of materials, namely as polymeric waste materials, organic waste materials, glass waste materials, etc., are not pure, nor of a narrow diversity. Each or some of the groups of waste materials, e.g., polymeric waste materials, may comprise other waste components that are not of the same composition or class of materials, provided that the class of materials constitutes at least 75wt% of the waste material used. For example, a polymeric waste material may comprise at least 75wt% (or between 75 and 100wt%) of polymeric materials and may also comprise other materials, some of which may not be identified in the waste mass. Typically, each group or class of a waste material comprises between 75 and 100 wt% of the identified material, in different combinations of materials. For example, a polymeric waste material may comprise between 75 and 100 wt% of a mixture or a combination of different polymeric materials.

A process of the invention comprises treating a first stream of a first waste material under conditions converting the first waste material into a melt form, combining the melt with a second waste material (from a second stream of a second waste material) to obtain a homogenous mixture of the first and second waste materials, and combining the homogeneous mixture with one or more further (same or different) waste materials (from further streams of further and/or different waste materials) to obtain a homogenous melt, wherein optionally each of the first, second and further waste materials or streams is provided at a different temperature, and wherein the homogenous melt is processable into solid composite materials or objects.

A further process is provided for manufacturing a metal-free composite material from a waste material, the process comprising forming a homogenous melt of a waste material comprising a polymeric waste material, a glass-based waste material, a paper-based, a fabric waste material, and an organic-based waste material, and treating said melt with at least one additive, such as an organic additive, under conditions causing the melt (or extrudate) to form into a homogenous solid composite material.

Further provided is a process for manufacturing a metal-free composite material from a waste material, the process comprising:

-forming a homogenous melt of a waste material comprising a polymeric waste material, and/or a glass-based waste material, and/or a paper-based waste material, and/or a fabric waste material, and an organic-based waste material, and

-treating the melt with at least one organic additive under conditions causing the melt to form into a homogenous solid composite material.

A ***homogenous melt*** is obtained by continuously mixing the components of the various streams under conditions which provide an ideal or improved distribution of the materials in the melt. The homogenous melt lacks any significant amount of a material or a composition, such that the melt, extrudate or final product does not include a matter that is not evenly dispersed. In other words, the melt or product formed thereof has a uniform appearance or composition throughout.

In some embodiments, the forming of the homogenous melt comprises:

-treating a polymeric waste material and/or a glass waste material under conditions permitting formation of a homogeneous polymeric and/or glass melt, respectively;

-optionally treating said homogeneous polymeric and/or glass melt with paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the melt; and

-treating the melt with an organic-based waste material.

In some embodiments, the forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a homogeneous polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt; and

-optionally treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt.

In some embodiments, the forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a homogeneous polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt; and

-treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt.

In some embodiments, the process comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt;

-optionally treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt; and

-treating the mass or melt with at least one organic additive under conditions causing the mass or melt to form into a homogenous solid composite material.

In some embodiments, the forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt;

-treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt; and

-treating the mass or melt with at least one organic additive under conditions causing the mass or melt to form into a homogenous solid composite material.

In some embodiments, the process comprises treating a polymer/glass mass or melt with a shredded tire material. In some embodiments, shredded tires are added together with the at least one organic additive.

In some embodiments, the polymeric waste material is free of shredded tire material.

The process of the invention is staged such that each of the steps of the process may be carried out at continuously lower temperatures enabling, on one hand, addition of thermally unstable components, such as organic waste materials and additives, and on the other hand, preventing the melt or mass from solidifying before a homogenous mixture of waste materials is obtained or before the process is completed.

In a first step of the process, a polymeric waste and/or a glass waste is treated under conditions which transform the waste mass into a melt, without causing decomposition and/or taring or the polymeric mass.

In some embodiments, a polymeric waste is treated in the first step. As the polymeric waste may comprise a variety of unsorted polymeric materials, some having melting temperatures higher than others, the conditions selected for achieving the melt allow for melting of some of the polymeric waste and dissolution of some of the other polymeric waste in the melt; thereby a achieving a homogenous polymeric melt. The conditions employed may involve heating the polymeric waste at a temperature between 180 and 250°C, while mixing or blending the materials together, e.g., at ambient pressures.

The temperature used for melting the polymeric waste may be not greater than 250°C. In some embodiments, the temperature is between 180 and 250°C, between 180 and 240°C, between 180 and 230°C, between 180 and 220°C, between 180 and 200°C, between 190 and 250°C, between 190 and 240°C, between 190 and 230°C, between 190 and 220°C, between 200 and 250°C, between 200 and 240°C, between 200 and 230°C, or between 200 and 220°C.

Where glass waste is treated first, the glass waste may be similarly treated. In some embodiments, however, the polymeric waste and the glass waste may be combined and treated together.

Alternatively, once the polymeric melt is obtained, a glass-based waste may be introduced into the polymeric melt maintained at a temperature between 180 and 250°C, as disclosed herein. To prevent the temperature of the melt from decreasing following the addition of the glass waste, the glass waste may be pre-treated to a temperature between 100°C and the temperature of the melt, being between 180 and 250°C. In other words, the glass waste may be pre-treated to a temperature between 100 and 250°C. In some embodiments, the temperature is between 100 and 240°C, or between 100 and 230°C, 100 and 220°C, 100 and 210°C, 100 and 200°C, 100 and 190°C, 100 and 180°C, 100 and 170°C, 100 and 160°C, 100 and 150°C, 100 and 140°C, 100 and 130°C, 100 and 120°C, 120 and 240°C, 130 and 240°C, 140 and 240°C, 150 and 240°C, 160 and 240°C, 170 and 240°C, 180 and 240°C, 190 and 240°C, 200 and 240°C, 110 and 160°C, 110 and 150°C, 120 and 160°C, 120 and 150°C, 130 and 160°C, or between 130 and 150°C.

To permit good thermal conduction throughout the glass mass, the glass waste may be premixed with components of the organic waste, such as oils, or with any oil waste from municipal or industrial sources, including non-food oils. The oils mixed with the glass waste may be any type of oil and may or may not contain soluble organic or inorganic materials. Thus, in a process of the invention, a glass-oil mixture is preformed and treated first under the aforementioned conditions; or may be premixed with the polymeric waste to provide a polymer/glass/oil mixture; or may be introduced into the polymeric melt, as disclosed.

Thus, in some embodiments, the step of "*treating a polymeric waste material and/or a glass waste material under conditions permitting formation of a homogeneous polymeric and/or glass melt, respectively*" comprises:

-treating a polymeric waste material under the conditions herein permitting formation of a polymeric melt; or

-treating a glass waste material under conditions permitting formation of a glass melt, wherein the glass waste is optionally provided in combination with oil waste; or

-treating a combination of a polymeric waste material and a glass waste material under conditions herein permitting formation of a homogeneous polymeric/glass melt, wherein the glass waste material may or may not be provided in combination with an oil waste (in which case the melt may be a polymer/glass/oil melt); or

-treating a polymeric waste material under conditions herein permitting formation of a polymeric melt, and adding thereinto a mixture of glass waste and oil, as disclosed herein.

Any of the above options constitutes an independent embodiment of a process of the invention.

In some embodiments, glass contained in any of the melts, e.g., polymeric/glass melt, or polymer/glass/oil melt, may be in a melted form, a semi-melted form or may remain in a solid form. The form may depend, *inter alia*, on the temperature utilized and the type of glass present.

In some cases, the glass waste material may be premixed with construction waste or building material waste derived from construction sites, demolition sites, or generally from the construction industry. In the alternative, the construction waste may be added into the polymeric melt prior to or after the addition of the glass waste (e.g., and oil waste). In other words, the step of "*treating a polymeric waste material and/or a glass waste*

material under conditions permitting formation of a homogeneous polymeric and/or glass melt, respectively" may comprise:

-treating a polymeric waste material under the conditions herein permitting formation of a polymeric melt, followed by adding thereinto a mixture of glass waste and optionally oil waste, optionally in combination with construction waste; or

-treating a glass waste material under conditions permitting formation of a glass melt, wherein the glass waste is optionally provided in combination with oil waste, followed by adding thereinto (or combining same with) a polymeric waste; or

-treating a combination of a polymeric waste material and a glass waste material under conditions herein permitting formation of a homogeneous polymeric/glass melt, wherein the glass waste material may or may not be provided in combination with an oil waste (in which case the melt may be a polymer/glass/oil melt), wherein construction waste may be separately added or provided admixed with the glass waste; or

-treating a polymeric waste material under conditions herein permitting formation of a polymeric melt, and adding thereinto a mixture of glass waste, oil waste and construction waste, as disclosed herein.

The polymeric melt or the polymer/glass melt or the polymer/glass/oil melt optionally comprising also construction waste, may be subsequently treated with waste derived from the electronic industry.

A first stage of the process thus comprises treating a polymeric waste material alone or in combination with:

-a glass waste material;

-a mixture of a glass waste material and oil waste;

-a construction waste material;

-a mixture of glass waste material, oil waste and construction waste material

at a temperature between 100 and 250°C, as disclosed herein, while mixing, permitting formation of a homogeneous melt.

In the next stage of the process, thermally sensitive waste components may be added. Before addition of the thermally sensitive waste materials, and depending on the temperature of the melt, the temperature of the process may be reduced to a temperature below a predetermined temperature at which the thermally sensitive materials decompose or tar or can undergo any unfavorable process. Typically, the temperature may be reduced to a maximum temperature of 200°C. Thus, the temperature of the melt is reduced to a

temperature between 150 and 200°C. In cases the temperature of the melt is already between 150 and 200°C, the temperature may not be reduced, unless necessary. At any event, a reduction in the temperature of the melt should not be to such a degree that solidification of the melt occurs.

At a temperature between 150 and 200°C paper-based waste, fabrics and textile, as well as wood waste products and cellulosic materials may be introduced into the polymeric melt, the polymer/glass melt or the polymer/glass/oil melt, as may be the case. In some cases, such paper-based waste products, fabrics or textiles are not present in the waste.

While maintaining the melt at a temperature not exceeding 200°C and following the addition of paper-based waste material, at least one organic additive may be added. At this stage of the process, shredded tires may also be introduced. Once added, the at least one organic additive, despite the relatively small amount added, i.e., between 0.5 and 2wt%, coagulates or amalgamates the melt mass, absorbs any of the oil components present in the melt, into a flowable hydrophobic material having, e.g., a viscosity ranging between 10,000 and 250,000 cP. The flowable material or extrudate may be subsequently cooled down in a mold or during shaping and structuring to obtain a hard molded or shaped solid composite, or cooled down and formed into aggregates for use as an additive to a variety of constructions elements.

As stated herein, the various waste materials may or may not be pretreated. For achieving a more effective process, the waste materials are typically shredded or mechanically treated into fragments of 1 to 10 mm. Larger or smaller fragments or segments may be used.

In a typical process of the invention, the relative amounts of the various waste components may be varied to allow a variety of products of different mechanical, chemical or physical properties. The relative amounts of waste materials may be:

Polymer waste material in an amount ranging between 10 and 76wt%; in some embodiments, the amount of the polymer materials in the composition may be between 10 and 75, 10 and 70, 10 and 65, 10 and 60, 10 and 55, 10 and 50, 10 and 45, 10 and 40, 10 and 35, 10 and 30, 20 and 75, 20 and 60, 20 and 50, 20 and 40, 30 and 75, 30 and 60, 30 and 50, 40 and 75, 40 and 65, 40 and 55, 50 and 75, or between 50 and 65 wt%;

Glass waste material in an amount ranging between 10 and 65wt%; in some embodiments, the amount of the glass materials in the composition may be between 20

and 65, 20 and 60, 20 and 55, 20 and 50, 20 and 45, 20 and 40, 20 and 35, 30 and 65, 30 and 60, 30 and 50, 30 and 40, 40 and 65, 40 and 60, or between 50 and 65wt%;

Oil material in an amount ranging between 4 and 20wt%; in some embodiments, the amount of the oil may be between 4 and 20, 4 and 15, 4 and 10, 4 and 9, 4 and 8, 4 and 7, 4 and 6, 8 and 20, 8 and 15, 10 and 20, or between 10 and 15wt%;

Construction waste material in an amount ranging between 25 and 65 wt%; in some embodiments, the amount of the construction waste material may be between 25 and 65, 25 and 60, 25 and 55, 25 and 50, 25 and 45, 25 and 40, 25 and 35, 30 and 65, 30 and 60, 30 and 50, 30 and 40, 40 and 65, 40 and 60, or between 50 and 65 wt%;

Paper-based material in an amount ranging between 0 and 45wt%, wherein in some embodiments, the composition is free of paper-based materials; in other embodiments, the amount of the paper-based material may be between 1 and 45wt%, or between 1 and 40, 1 and 35, 1 and 30, 1 and 25, 1 and 20, 1 and 15, 1 and 10, 5 and 40, 5 and 35, 5 and 30, 5 and 25, 5 and 20, 5 and 10, 10 and 40, 10 and 30, or between 10 and 20wt%;

Tires or polymers derived therefrom in an amount ranging between 0 and 55 wt%, wherein in some embodiments, the composition may be free of tires or processes tires; or may comprise tires or polymers derived therein in an amount between 1 and 55wt% or in amount between 1 and 55, 1 and 50, 1 and 45, 1 and 40, 1 and 35, 1 and 30, 1 and 25, 1 and 20, 1 and 15, 1 and 10, 1 and 9, 1 and 8, 1 and 7, 1 and 6, 1 and 5, 5 and 55, 5 and 50, 5 and 40, 5 and 30, 5 and 20, 5 and 10, 10 and 55, 10 and 45, 10 and 35, 10 and 25, 10 and 15, 20 and 55, 20 and 45, 20 and 35, 30 and 55, 30 and 45, or between 40 and 55wt%;

Additive, being optionally an organic additive, in an amount ranging between 0.5 and 2%, as disclosed herein.

The composition of waste materials may comprise additional unsorted materials. In the above, any composition is a combination of materials having a total weight amount of 100%.

The relative weight amounts of each the waste components in a composition may vary. The final amount of each component and/or a ratio between any two components present in the composition may be selected to modulate one or more mechanical, chemical or visual property of a final product made therefrom. An exemplary composition of waste materials is provided in **Table 7**.

No.	Material	Relative weight (wt%)
1	Plastic or polymeric material	30-60
2	oil	10-20
3	organic additive	0.5-2
4	Colored plastic	0.1-4
5	black plastic	9-20
6	metal-free cement	20-40
7	fabrics or textiles	5-15
8	paper products	5-15
9	tires	50-55

Table 7: Relative weight amounts of material components in compositions of the invention

The extrudate or melt obtained of a process of the invention, may be treated under *conditions causing the mass or melt to form into a homogenous solid composite material*. These conditions involve the staged temperatures, the continuous mixing and the addition of the at least one additive, being at least one organic additive. The additive may be generally selected as above.

In some embodiments, the additive is selected from (i) yeasts, such as of the species *Saccharomyces cerevisiae*, *Saccharomyces exiguus* (*S. minor*) or others; (ii) carbohydrates, such as glucose, galactose, fructose, xylose, sucrose, lactose, maltose, isomaltulose, trehalose, sorbitol, mannitol; oligosaccharides such as maltodextrins, raffinose, stachyose, fructo-oligosaccharides and others; polysaccharides such as starch, amylose, amylopectin, modified starches, glycogen, cellulose, hemicellulose, pectins, hydrocolloids, modified starch such as acid-treated starch, alkaline-treated starch, bleached starch, oxidized starch, enzyme-treated starch, mono-starch phosphate, di-starch phosphate, phosphated di-starch phosphate, acetylated di-starch phosphate, starch acetate, acetylated di-starch adipate, hydroxypropyl starch, hydroxypropyl di-starch phosphate, hydroxypropyl di-starch glycerol, starch sodium octenyl succinate, acetylated oxidized starch, cationic starches, hydroxyethyl starch, carboxymethylated starches and others, (iii) natural gums, such as alginic acid, beta-glucan, caranna, chicle, dammar gum, galactomannan, gellan gum, glucomannan, guar gum, gum anima, gum Arabic, gum guaicum, gum karaya, konjac, locust bean gum, mastic, myrrh, neem gum, psyllium,

tragacanth, welan gum, xanthan gum and others, (iv) clay, talc, dextrin, maltodextrin, starch, textured vegetable proteins, casein, proteins, flour, corn flour and others.

In some embodiments, the at least one additive is at least one gum or a combination of two or more gums.

In some embodiments, the at least one additive is at least one gum and at least one carbohydrate.

In some embodiments, the at least one additive is yeast and at least one gum.

Presence of the at least one additive in a small amount of between 0.2 and 5wt% allows for the manufacturing of a stable, homogenous pores-free material which has the properties disclosed herein. In the absence of the at least one additive, e.g., at least one gum, such properties are not obtained or are less improved.

Solid composite formed by processes of the invention may be generally used in two forms: as a final product following molding or structuring of the hot melt or hot extrudate, or as an additive or a reinforcing material in methods of production of other products. The hot melt may be molded or shaped into a variety of objects or elements suitable for use as construction elements or as ornamental objects. These objects and elements may be pots, containers, roof tiles, slabs, columns, walls, beams, blocks, bricks, ingots, panels, paver units, stairs, tiles, flags, and concrete masonry units (CMU) which may be manufactured as individual elements or as a plurality of elements, or an assembly of elements.

The hot melt may additionally be cooled down to provide a solid composite mass of an amorphous form that can be shredded or cut or milled or otherwise mechanically treated to provide particles or fragments or stones or a composite material that may be added to a plurality of other construction components to manufacture a reinforced element, wherein such components may be concrete, formwork, and steel.

Thus, the invention further provides a solid object formed of a melt composite manufactured according to the invention. In some embodiments, the solid object is of any shape and size, which may be provided as a single functional unit or a unit suitable for assembly with one or more identical or different elements. The object may be a pot, a container, a roof tile, a slab, a column, a construction beam, a wall, a block, a brick , a panel, a paver unit, a stair, a tile, a flag, and a concrete masonry unit (CMU).

The invention further provides a construction component or a cement additive, the component or cement additive being a composite formed according to the invention,

wherein the composite is in a form of powder, a granulated material, or a particulate material (nano, micro or macro in size).

In some embodiments, the construction component or cement additive is suitable as an additive for any cement composition, wherein the cement may be selected from pastes, mortars, and concrete compositions, which may be based on Ordinary Portland Cement as a hydratable binder.

The invention further provides concrete composition comprising cement and a granulated material being a particulate material of the invention. In some embodiments, the particulate material may be provided in combination with fine aggregates, such as sand, or in place of coarse aggregates such as crushed gravel and stones.

Products obtained from processes of the invention vary in composition and properties, mainly due to variations in the waste compositions. The variations may, however, be minimized due to the selective stepwise addition of the waste materials. Common to all products of the invention are superior properties, which exceed properties known for cementitious material and concrete-based objects. Objects of the invention exhibit one or more of the following properties:

-objects do not absorb water,

-objects do not suffer from water-induced degradation;

-objects are water repellents;

-objects are hydrophobic;

-objects are oil repellents;

-objects are stain resistant;

-objects are shock absorbers;

-objects exhibit compression strengths of 60MPa (as compared to 24MPa for concrete objects);

-objects are thermally stable at temperatures up to 600°C;

-objects are thermal insulators;

-objects are resistant to strong acids;

-objects are non-absorbing;

-objects are stain resistant;

-objects are acoustic materials capable of reducing noise generated on both sides thereof; etc.

Objects of the invention may be characterized by one or more of the following mechanical properties:

-Flexural Modulus (yield, at 25°C) – about 20MPa;

-Tensile Strength (yield, at 25°C) – about 30Mpa;

-Elongation (yield, at 25°C) – between about 0.5 and 18%;

-Shore Hardness, D Scale – 50-90;

-Continues service temperature: between -70 (minus 70) and 80°C;

-Water absorption (24hr. immersion) of between 0.01~0.03%; and

-Density: between 0.67 and 1.2gr/cm^3;

The invention further provides a waste conversion system, the system comprising

-at least one first unit for receiving a stream of a first waste material into a heated reaction chamber, the heated reaction chamber being configured to permit continuous mixing of its content at a temperature suitable for converting the first waste material into a form of a melt;

-at least one second heated unit for receiving and modifying temperature of a stream of a second waste material, the at least one second heated unit being configured to feed the second waste material into the heated reaction chamber to mix the melt of the first waste material with the heated second waste material to provide a second melt;

-at least one further heated unit for receiving a stream of a further waste material and feeding said further waste material into the heated reaction chamber comprising the second melt; and

-a nozzle assembly or an output valve configured and operable to output a combined melt.

In a system of the invention, multiple material streams may be each flown or delivered or carried into or directed into a different unit for receiving the waste stream, wherein one or more of the units is provided with a heating surface or heating unit which purpose is to modify a temperature of the stream. Typically, modifying the temperature of the stream includes a provision of heating or raising the temperature of the stream to a predetermined temperature. Each waste stream may be fed into a separate unit or several different streams may be fed into a single unit.

The heated reaction chamber may be an extruder provided with wo or more feeding units or units for receiving streams of waste materials, such that each of the streams is fed into the extruder under different conditions and in sequence. Thus, the

invention further provides a waste conversion system, the system comprising a single screw heated extruder, the extruder being fitted with a plurality of receiving units or hoppers for receiving a plurality of waste material fractions or streams, wherein

-a first unit of the plurality of receiving units is configured for receiving a stream of a first waste material into the heated extruder operable at a temperature suitable for converting the first waste material into a form of a melt;

-a second heated unit of the plurality of receiving units positioned downstream to the first unit for receiving and heating a stream of a second waste material, the second unit being configured to feed the heated second waste material into the extruder to mix the melt of the first waste material with the heated second waste material to form into a second melt;

-a further heated unit of the plurality of receiving units positioned downstream to the second heated unit for receiving a stream of a further waste material and feeding said heated further waste material into the extruder to form a further melt; and

-a nozzle assembly or a die or an output valve configured downstream to the further heated unit and operable to output the further melt.

A system of the invention, being in some configurations a single screw extruder, comprises an elongated barrel that is equipped with a screw which conveys the melt materials along the barrel (downstream) from the first feeding unit and then pushes the final melt, i.e., the further melt through the die. The screw is rotated at a predetermined speed, operated by an electric motor drive unit and gearbox. Heating elements, controlled by temperature controlling units, are positioned on the barrel circumference to maintain the temperature at a set-point temperature. Each of the receiving units may also be provided with one or more heating elements to pre-heat the waste material prior to feeding into the reaction chamber or the extruder.

As noted herein, a process of the invention is staged such that each of the steps of the process may be carried out at continuously lower temperatures. Thus, each of the receiving units or different sections of the reaction chamber or extruder may be provided at a different temperature. A first region of the extruder may extend between the first and second receiving units, a second region of the extruder may extend between the second and the further receiving units and a third or further region of the extruder may extend the distance between the further receiving unit and the output valve or die, such that each extruder region is maintained or operated at a different temperature. In some

configurations, the temperature of the first extruder region is the highest and the temperature of the further region is the lowest. In some embodiments, the temperature difference between any two extruder regions is between 50º and 100°C.

In some embodiments, the first receiving unit is provided unheated, while the first region of the extruder receiving the waste stream may be heated to a temperature between 180 and 250°C. The second and further receiving units may be heated to a temperature between 100 and 200°C and each of the second and further extruder regions may be maintained or operated at a temperature below 180°C or between 100 and 180°C.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments will now be described, by way of non-limiting example only, with reference to the accompanying drawings, in which:

Fig. 1 provides a schematic depiction of a process according to certain embodiments of the invention.

Fig. 2 provides a schematic depiction of a further process according to embodiments of the invention.

Fig. 3 provides a schematic depiction of yet another process according to embodiments of the invention.

Fig. 4 provides a schematic depiction of a thermal profile of a process according to embodiments of the invention.

Fig. 5 provides a schematic depiction of a system according to embodiments of the invention.

Figs. 6A-B provide images of aggregates formed according to a process according to embodiments of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

As disclosed herein, objects of the invention are manufactured from a mixture of unsorted waste materials. Thermal treatment of compositions of matter according to the present disclosure, resulted in hard composite materials that can be shaped into a variety of end products.

In most general terms and as generally depicted in **Fig. 1**, a process of the invention **10** comprises treating a first stream of a first waste material **12** under conditions **14** capable of converting the first waste material **12** into a melt form **16**. The melt **16** is then combined with a second waste material **18** provided by a second material stream to obtain a homogenous mixture **20** of the first and second waste materials. Mixture **20** is combined with one or more further different waste materials **22a**, **22b**, **22c**, etc and at least one organic additive **24**, to obtain a homogenous melt **26**, which may be processed into solid composite materials or objects or products **30**.

In the process configuration depicted in **Fig. 2**, a homogenous melt **40** (block 26 shown in **Fig. 1**) is formed by treating a polymeric waste material **12** and/or a glass waste material **18** under conditions **12a** or **18a** permitting formation of a homogeneous polymeric melt and/or glass melt **32a** and **32b**, respectively; optionally treating said homogeneous polymeric and/or glass melt **32a/32b** with paper-based and/or fabric waste material **34** under conditions **34a** or **34b** permitting incorporation of the paper-based and/or fabric waste material **34** in the melt **32a/32b** to obtain an incorporated melt **36a** or **36b**; and treating the melt **36a/36b** with an organic-based waste material **38**, which may be an additive such as an organic additive.

The configuration of **Fig. 2** can be carried out as depicted in **Fig. 3** by treating a polymeric waste material **12** under conditions **12a** permitting formation of a polymeric melt **32**; treating said polymeric melt **32** with a glass waste material **18**, allowing said melt **32** to mix with the glass waste material **18** to obtain a homogenous polymer/glass mass or melt **42**; and optionally treating said homogeneous polymer/glass mass or melt **42** with a paper-based and/or fabric waste material **44** under conditions **44a** permitting incorporation of the paper-based and/or fabric waste material **44** in the mass or melt to obtain an incorporated melt **46**. The incorporated melt **46** or the polymer/glass melt **42** may be further treated with at least one organic additive **48** under conditions **50a** or **50b**, causing the mass or melt **46** to form into a homogenous solid composite material **52**.

As further depicted in **Fig. 3**, the polymer/glass mass or melt **42** may be treated with a shredded tire material **54**.

As disclosed herein, the conditions under which the melt forms are obtained in each of the processes of the invention, include thermal treatment of the waste combinations, under continuous mixing. The thermal conditions used in the process stages are depicted in **Fig. 4**.

As depicted in **Fig. 4**, a process of the invention is carried out under a continuously reduced temperature profile, such that heat-sensitive materials, such as the organic waste and organic additive, are introduced at a lower temperature to the temperature of the heat-stable waste materials, e.g., construction waste and polymeric waste. The thermal profile is further reflected in the structuring of systems of the invention, as exemplified in **Fig. 5**.

Fig. 5 depicts a general system for converting a waste material into a solid product or object or material. A system **100** includes at least one first unit **110** for receiving a stream **110a** of a first waste material into a heated reaction chamber **105**, being designed, in some configurations as an extruder barrel; the heated reaction chamber **105** being configured to permit continuous mixing of its content at a temperature suitable for converting the first waste material into a form of a melt; at least one second heated unit **120** for receiving and modifying temperature of a stream **120a** of a second waste material is configured to feed a second waste material into the heated reaction chamber **105** to mix the melt of the first waste material with the heated second waste material to provide a second melt; at least one further heated unit **130** for receiving a stream **130a** of a further waste material and feeding said further waste material into the heated reaction chamber **105** comprising the second melt; and a nozzle assembly or an output valve **140** configured and operable to output a combined melt or an extrudate.

In some configurations, the system **100** may be an extruder system, comprising a single screw **150**, the extruder being fitted with a plurality of receiving units or hoppers **110**, **120**, **130**...for receiving a plurality of waste material fractions or streams **110a**, **120a**, **130a**..., as shown in **Fig. 5**. The system reaction chamber **105** may be equipped with a plurality of heating elements **155**, **156**, positioned on a barrel circumference to maintain the temperature at a set-point temperature. Each of the receiving units **110**, **120**, **130** may be provided with one or more heating elements **110b**, **120b**, **130b**... to pre-heat the waste material prior to feeding into the reaction chamber or the extruder **105**.

The system **100** may be defined as having a plurality of regions maintained at different temperatures. A first region **200** extending between the first and second receiving units; a second region **300** extending between the second and the further receiving units; and a third or further region **400** extending the distance between the further receiving unit and the output valve or die, such that each region is maintained or operated at a different temperature, wherein the temperature difference between any two

regions is between 50° and 100°C. In the particular configuration, the first receiving unit **110** is provided unheated and the first region **200** receiving the waste stream is maintained at a temperature between 180 and 250°C. The second **120** and further **130** receiving units are each heated to a temperature between 100 and 200°C and each of the second **300** and further **400** regions is maintained or operated at a temperature below 180°C or between 100 and 180°C.

Products of the invention are unique solid materials which exhibit improved mechanical and physical properties as compared to cementitious products or stone-based products. A solid mass manufactured from a waste material with an organic additive such as a natural gum, being one or a combination of beta-glucan, gellan gum, guar gum, gum Arabic, mastic, and xanthan gum was tested in a variety of ways to measure its mechanical and physical properties. Aggregates formed by mechanically breaking the solid product demonstrated material continuity (**Figs. 6A-B**), with minimal surface porosity rendering the surface stain resistant, water resistant and resistance to acidic and basic environments.

The aggregates were found to be 50-70% lighter than traditional aggregates used in constructions.

Also, a 20X10X6 cm brick was manufactured and was tested to measure water absorptivity, stain resistance and acid degradability. Water, salt water, ketchup, oil, coffee, red wine, sodium chlorate, citric acid, inks and a variety of food products were used to stain or cover a surface region of the brick for a period of up to 24 hours. Then the materials were wiped and the surface of the brick was tested. No water absorption or staining or degradation were observed. Same tests were applied to a conventional cement-based brick used for paving. The cemented-based brick absorbed the water and was stained to a degree where the stain could not be removed, potentially due to absorption of the stating material into the cementitious material.

A 10x10x6 brick was pressed to break and was compared to a cement-based brick of substantially similar dimensions. While a brick manufactured according to the invention sustained a pressure of at least 9 tons, the cement-based brick was fractured at a pressure of 3.5 tons.

The above data supports the uniqueness and superiority of products of the invention over commercially available cementitious products and other solid products formed from waste materials.

CLAIMS:

1. A process for forming a 3D object form a metal-free waste material, the process comprising treating a first stream of a first waste material under conditions converting the first waste material into a melt form, combining the melt with a second stream of a second waste material to obtain a homogenous mixture of the first and second waste materials, and combining the mixture with one or more further streams of further and different waste materials and at least one organic additive, to obtain a homogenous melt, wherein each of the first, second and further streams is provided at a different temperature, and wherein the homogenous melt is processable into solid composite materials or objects.

2. A process for manufacturing a metal-free composite material from a waste material, the process comprising:

-forming a melt of a polymeric waste material, and/or a glass-based waste material, and/or a paper-based waste material, and/or a fabric waste material, and an organic-based waste material, and

-treating the melt with at least one organic additive under conditions causing the melt to form into a homogenous solid composite material.

3. The process according to claim 1 or 2, wherein forming of the homogenous melt comprises:

-treating a polymeric waste material and/or a glass waste material under conditions permitting formation of a polymeric and/or glass melt, respectively;

-optionally treating said polymeric and/or glass melt with paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the melt; and

-treating the melt with an organic-based waste material.

4. The process according to claim 1 or 2, wherein forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt; and

-optionally treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt.

5. The process according to claim 1 or 2, wherein forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt; and

-treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt.

6. The process according to claim 1 or 2, wherein the process comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt;

-optionally treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt; and

-treating the mass or melt with at least one organic additive under conditions causing the mass or melt to form into a homogenous solid composite material.

7. The process according to claim 1 or 2, wherein forming of the homogenous melt comprises:

-treating a polymeric waste material under conditions permitting formation of a polymeric melt;

-treating said polymeric melt with a glass waste material, allowing said melt to mix with the glass waste material to obtain a homogenous polymer/glass mass or melt;

-treating said homogeneous polymer/glass mass or melt with a paper-based and/or fabric waste material under conditions permitting incorporation of the paper-based and/or fabric waste material in the mass or melt; and

-treating the mass or melt with at least one organic additive under conditions causing the mass or melt to form into a homogenous solid composite material.

8. The process according to any one of the preceding claims, the process comprising treating a polymer/glass mass or melt with a shredded tire material.

9. The process according to claim 8, wherein the shredded tires are added together with the at least one organic additive.

10. The process according to any one of the preceding claims, wherein the polymeric waste material is free of shredded tire material.

11. The process according to any one of claims 3 to 10, wherein the homogeneous polymeric and/or glass melt is obtained at a temperature between 180 and 250°C, while mixing or blending.

12. The process according to any one of claims 1 to 11, wherein the glass waste is premixed with at least one component of an organic waste.

13. The process according to any one of claims 1 to 11, wherein the glass waste is premixed with at least one oil waste.

14. The process according to claim 3, wherein the treating of the polymeric waste material and/or the glass waste material under conditions permitting formation of a polymeric and/or glass melt comprises:

-treating a polymeric waste material under the conditions permitting formation of a polymeric melt; or

-treating a glass waste material under conditions permitting formation of a homogeneous glass melt, wherein the glass waste is provided in combination with oil waste; or

-treating a combination of a polymeric waste material and a glass waste material under conditions permitting formation of a homogeneous polymeric/glass melt, wherein the glass waste material is optionally provided in combination with an oil waste; or

-treating a polymeric waste material under conditions permitting formation of a homogeneous polymeric melt, and adding thereinto a mixture of glass waste and oil.

15. The process according to any one of the preceding claims, wherein the glass waste material is premixed with construction waste or building waste material.

16. The process according to any one of claims 3 to 15, wherein construction waste is premixed with the polymeric melt prior to or after addition of the glass waste and oil waste.

17. The process according to claim 3, wherein the treating of the polymeric waste material and/or the glass waste material under conditions permitting formation of a polymeric and/or glass melt comprises:

-treating a polymeric waste material under the conditions permitting formation of a polymeric melt, followed by adding thereinto a mixture of glass waste and oil waste, optionally in combination with construction waste; or

-treating a glass waste material under conditions permitting formation of a glass melt, wherein the glass waste is provided in combination with oil waste, followed by adding thereinto a polymeric waste; or

-treating a combination of a polymeric waste material and a glass waste material under conditions permitting formation of a homogeneous polymeric/glass melt, wherein the glass waste material is optionally provided in combination with an oil waste, wherein construction waste is separately added or provided admixed with the glass waste; or

-treating a polymeric waste material under conditions permitting formation of a polymeric melt, and adding thereinto a mixture of glass waste, oil waste and construction waste.

18. The process according to any one of the preceding claims, wherein paper-based waste, fabrics and textile, wood waste products and cellulosic materials are introduced into the polymeric melt, the polymer/glass melt or the polymer/glass/oil melt at a temperature between150 and 200°C.

19. The process according to any one of the preceding claims, wherein the at least one organic additive is added at a temperature below 200°C.

20. The process according to any one of the preceding claims, wherein the waste material is mechanically pre-treated to reduce the waste into fragments or particulates.

21. The process according to claim 20, wherein the fragments or particulates having an averaged size of between 1 to 10 mm.

22. The process according to any one of the preceding claims, wherein the polymeric waste comprises one or more of plastic, rubber, low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polyurethane, acrylonitrile butadiene styrene, cellulose acetate, cellulose acetate butyrate, ethylene-vinyl acetate, fluorinated ethylene propylene, fluoropolymers, high-performance plastics, Pearloid, perfluoroalkoxy alkane, perfluoroether, Plastisol, poly(ethyl methacrylate), poly(methyl methacrylate),

polyphenylene sulfide, polyacrylonitrile, polyamide, polyamide-imide, polyaryletherketone, polybutadiene, polybutylene succinate, polybutylene terephthalate, polycaprolactone, polycarbonate, polychlorotrifluoroethylene, polycyclohexylenedimethylene terephthalate, polyester, polyether ether ketone, polyetherimide, polyetherketoneketone, polyhydroxyalkanoates, polyhydroxybutyrate, polyimide, polyketone, polylactic acid, polymethylpentene, polyoxymethylene, polyphthalamide, polystyrene, polysulfone, polyvinyl ester, polyvinyl fluoride, polyvinylcarbazole, bakelite, epoxy, furan resins, melamine resins, Novolak, polybenzoxazine, polyhexahydrotriazine, polyisocyanurate, silicone, urea-formaldehyde, and vinyl ester resin.

23. The process according to any one of the preceding claims, wherein the amount of the polymeric waste is between 10 and 76wt%.

24. The process according to any one of the preceding claims, wherein the tires comprise metal-free tire fragments.

25. The process according to any one of the preceding claims, wherein the amount of the tires waste is between 0 and 55 wt%.

26. The process according to any one of the preceding claims, wherein the construction waste material comprises one or more non-metallic composite panels, artificial stone, asphalt, bricks and building stones, concrete, glass materials, insulation materials, polymeric materials, sand, soil-based materials, terracotta, and wood.

27. The process according to any one of the preceding claims, wherein the amount of the construction waste material is between 25 and 65 wt%.

28. The process according to any one of the preceding claims, wherein the glass waste material comprises glass and ceramic materials derived from construction materials, glass industry, municipal waste, optic systems, or glassware.

29. The process according to any one of the preceding claims, wherein the amount of the glass waste material is between 20 and 65wt%.

30. The process according to any one of the preceding claims, wherein the paper waste comprises one or more of books, newspapers, magazines, posters, wrapping materials, pamphlets, maps, signs, labels, advertisements, notebooks, writing pads, envelopes, packaging materials, stationery, parchment, toilet paper, paper towels, paper plates and paper cups, beverage cartons, tea bags, food packaging, coffee filters, and wallpaper.

31. The process according to any one of the preceding claims, wherein the amount of the paper waste material is between 0 and 45wt%.

32. The process according to any one of the preceding claims, wherein the at least one oil is an edible oil or an inedible oil.

33. The process according to claim 32, wherein the edible oil is selected from coconut oil, corn oil, canola oil, cottonseed oil, olive oil, palm oil, peanut oil, rapeseed oil, safflower oil, sesame oil, soybean oil, sunflower oil, almond oil, beech nut oil, Brazil nut oil, cashew oil, hazelnut oil, macadamia oil, mongongo nut oil, pecan oil, pine nut oil, pistachio oil, walnut oil, and pumpkin seed oil.

34. The process according to claim 32, wherein the inedible oil is selected from copaiba oil, jatropha oil, jojoba oil, milk bush, nahor oil, paradise oil, petroleum nut oil, pongamia oil, dammar oil, linseed oil, poppyseed oil, stillingia oil, tung oil, vernonia oil, karanja, mahua, rubber seed, cottonseed, neem, putranjiba, tobacco seed, polanga seed, cardoon, deccan hemp, castor, jojoba, moringa, cuphea, and poon.

35. The process according to claim 32, wherein the inedible oil is an industrial oil.

36. The process according to claim 35, wherein the industrial oil is selected from gear oils, air compressor oils, hydraulic oils, rock drill oils, turbine oils, engine oils, and lubricants.

37. The process according to any one of the preceding claims, wherein the amount of the at least one oil is between 4 and 20wt%.

38. The process according to any one of the preceding claims, wherein the at least one organic additive is selected to endow mechanical attributes to a manufactured product.

39. The process according to claim 38, wherein the at least one organic additive is selected from yeast, carbohydrates and natural gums.

40. The process according to claim 38 or 39, wherein the at least one organic additive is yeast.

41. The process according to claim 40, wherein the yeast is Saccharomyces cerevisiae or Saccharomyces exiguus (S. minor).

42. The process according to claim 38 or 39, wherein the art least one organic additive is a carbohydrate.

43. The process according to claim 42, wherein the carbohydrate is one or more of glucose, galactose, fructose, xylose, sucrose, lactose, maltose, isomaltulose, trehalose, sorbitol, mannitol; oligosaccharides such as maltodextrins, raffinose, stachyose, fructo-

oligosaccharides and others; polysaccharides such as starch, amylose, amylopectin, modified starches, glycogen, cellulose, hemicellulose, pectins, and hydrocolloids.

44. The process according to claim 43, wherein the at least one organic additive is selected from maltodextrin, dextrose, maltitol, erythritol, sorbitol, mannitol, and modified starch.

45. The process according to claim 44, wherein the modified starch is acid-treated starch, alkaline-treated starch, bleached starch, oxidized starch, enzyme-treated starch, mono-starch phosphate, di-starch phosphate, phosphated di-starch phosphate, acetylated di-starch phosphate, starch acetate, acetylated di-starch adipate, hydroxypropyl starch, hydroxypropyl di-starch phosphate, hydroxypropyl di-starch glycerol, starch sodium octenyl succinate, acetylated oxidized starch, cationic starches, hydroxyethyl starch, or carboxymethylated starch.

46. The process according to claim 38, wherein the at least one organic additive is a natural gum.

47. The process according to claim 46, wherein the natural gum is alginic acid, beta-glucan, caranna, chicle, dammar gum, galactomannan, gellan gum, glucomannan, guar gum, gum anima, gum Arabic, gum guaicum, gum karaya, konjac, locust bean gum, mastic, myrrh, neem gum, psyllium, tragacanth, welan gum, or xanthan gum.

48. The process according to claim 38, wherein the at least one organic additive is selected from clay, talc, dextrin, maltodextrin, starch, textured vegetable proteins, casein, proteins, flour, and corn flour.

49. The process according to any one of the preceding claims, wherein the waste material is free of metals and ash.

50. The process according to any one of the preceding claims, comprising molding or structuring the homogenous melt into a 3D object.

51. The process according to claim 50, wherein the 3D object is a construction element or an ornamental object.

52. The process according to claim 51, wherein the 3D object is selected amongst pots, containers, roof tiles, slabs, columns, walls, beams, blocks, bricks, ingots, panels, paver units, stairs, tiles, flags, and concrete masonry units (CMU).

53. The process according to any one of claims 1 to 51, comprising cooling the homogenous melt into a solid and mechanically transforming said solid into a particulate material.

54. The process according to claim 53, wherein the particulate material is for use as an additive for a cement composition, wherein the cement composition is selected from pastes, mortars, and concrete compositions.

55. A process for forming a 3D object form a metal-free waste material, the process comprising

-treating a first stream of a first waste material under conditions converting the first waste material into a melt form,

-combining the melt with a second stream of a second waste material to obtain a homogenous mixture of the first and second waste materials,

-combining the mixture with one or more further streams of further and different waste materials and further with at least one organic additive being yeast, a carbohydrate and/or a natural gum,

to obtain a homogenous melt convertible into a 3D object.

56. The process according to claim 55, wherein each of the process steps is carried out under different conditions.

57. The process according to claim 56, wherein the conditions converting the first waste material into a melt form include a temperature between 180 and 250°C, the conditions used to combine the mixture with one or more further streams include a temperature between 150 and 200°C, and the conditions for adding the at least one organic additive include a temperature up to 180°C.

58. A waste conversion system, the system comprising

-at least one first unit for receiving a stream of a first waste material into a heated reaction chamber, the heated reaction chamber being configured to permit continuous mixing of its content at a temperature suitable for converting the first waste material into a form of a melt;

-at least one second heated unit for receiving and modifying temperature of a stream of a second waste material, the at least one second heated unit being configured to feed the second waste material into the heated reaction chamber to mix the melt of the first waste material with the heated second waste material to provide a second melt;

-at least one further heated unit for receiving a stream of a further waste material and feeding said further waste material into the heated reaction chamber comprising the second melt; and

-a nozzle assembly or an output valve configured and operable to output a combined melt.

59. The system according to claim 58, being an extruder system.

60. A waste conversion system, the system comprising a single screw heated extruder, the extruder being fitted with a plurality of receiving units or hoppers for receiving a plurality of waste material fractions or streams, wherein

-a first unit of the plurality of receiving units is configured for receiving a stream of a first waste material into the heated extruder operable at a temperature suitable for converting the first waste material into a form of a melt;

-a second heated unit of the plurality of receiving units positioned downstream to the first unit for receiving and heating a stream of a second waste material, the second unit being configured to feed the heated second waste material into the extruder to mix the melt of the first waste material with the heated second waste material to form into a second melt;

-a further heated unit of the plurality of receiving units positioned downstream to the second heated unit for receiving a stream of a further waste material and feeding said heated further waste material into the extruder to form a further melt; and

-a nozzle assembly or a die or an output valve configured downstream to the further heated unit and operable to output the further melt.

61. The system according to claim 60, wherein the single screw extruder comprises an elongated barrel equipped with a screw conveying the melt materials along the barrel from the first feeding unit pushing the melt through the die.

62. The system according to claim 61, wherein the single screw is rotated at a predetermined speed, operated by an electric motor drive unit and gearbox.

63. The system according to any one of claims 58 to 62, equipped with a plurality of heating elements, controlled by one or more temperature controlling units, positioned on a barrel circumference to maintain the temperature at a set-point temperature.

64. The system according to any one of claims 58 to 63, wherein each of the receiving units is provided with one or more heating elements to pre-heat the waste material prior to feeding into the reaction chamber or the extruder.

65. The system according to any one of claims 58 to 64, having a first region extending between the first and second receiving units, a second region extending between the second and the further receiving units and a third or further region extending

the distance between the further receiving unit and the output valve or die, such that each region is maintained or operated at a different temperature, wherein the temperature of the first region being higher than the temperature of the second region and the further region.

66. The system according to claim 65, wherein the temperature difference between any two regions is between 50^0 and 100°C.

67. The system according to claim 65 or 66, wherein the first receiving unit is provided unheated and the first region receiving the waste stream is maintained at a temperature between 180 and 250°C.

68. The system according to any one of claims 65 to 67, wherein the second and further receiving units are each heated to a temperature between 100 and 200°C and each of the second and further regions is maintained or operated at a temperature below 180°C or between 100 and 180°C.

69. An object formed of a process according to any one of claims 1 to 57.

70. The object according to claim 69 being in an aggregate form.

71. Aggregates formed of a process according to any one of claims 1 to 57.

72. A construction unit formed of or comprising aggregates according to claim 71.

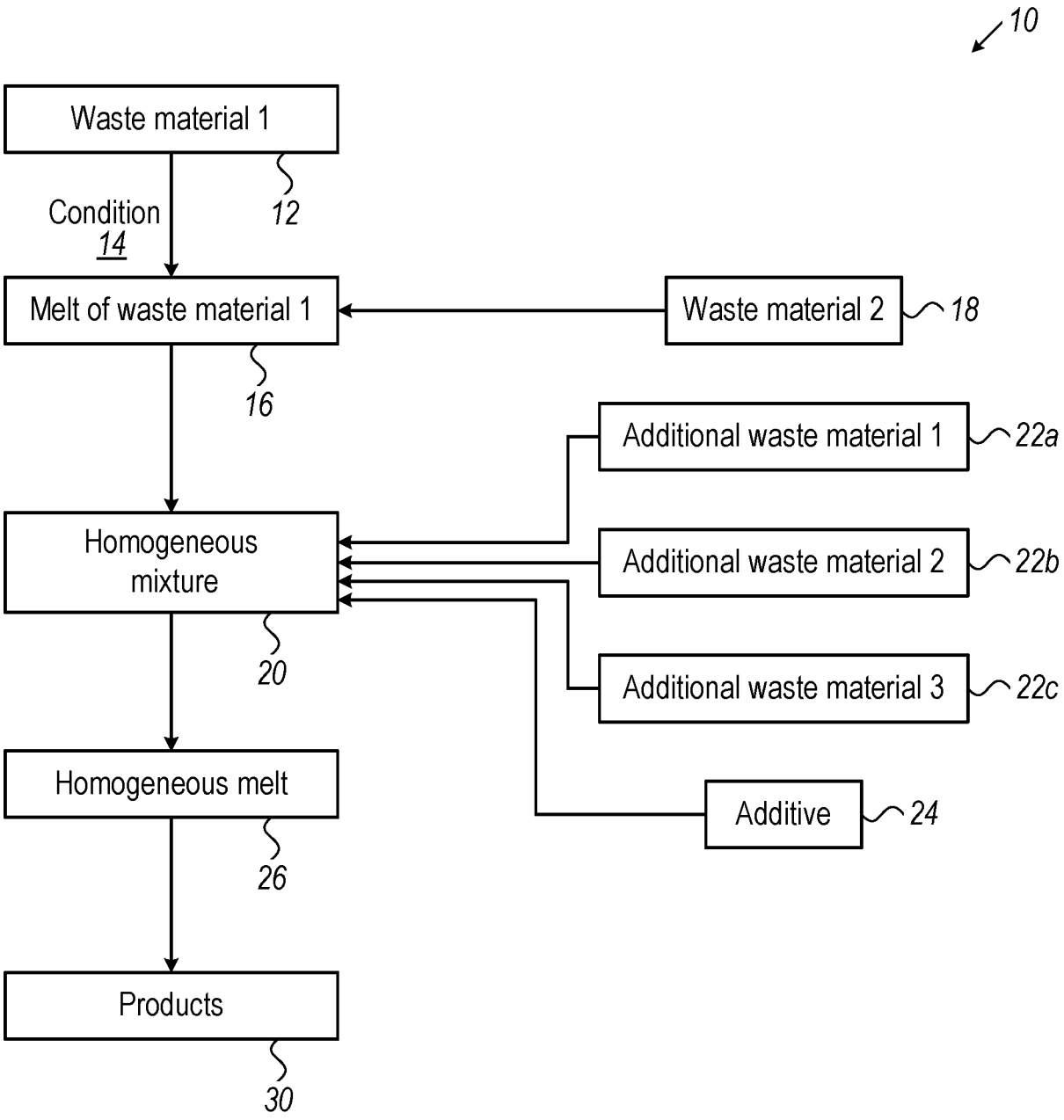

10
Waste material 1
12
Condition
14
Melt of waste material 1
Waste material 2
18
16
Additional waste material 1
22a
Homogeneous
mixture
Additional waste material 2
22b
20
Additional waste material 3
22c
Homogeneous melt
Additive
24
26
Products
30

FIG. 1

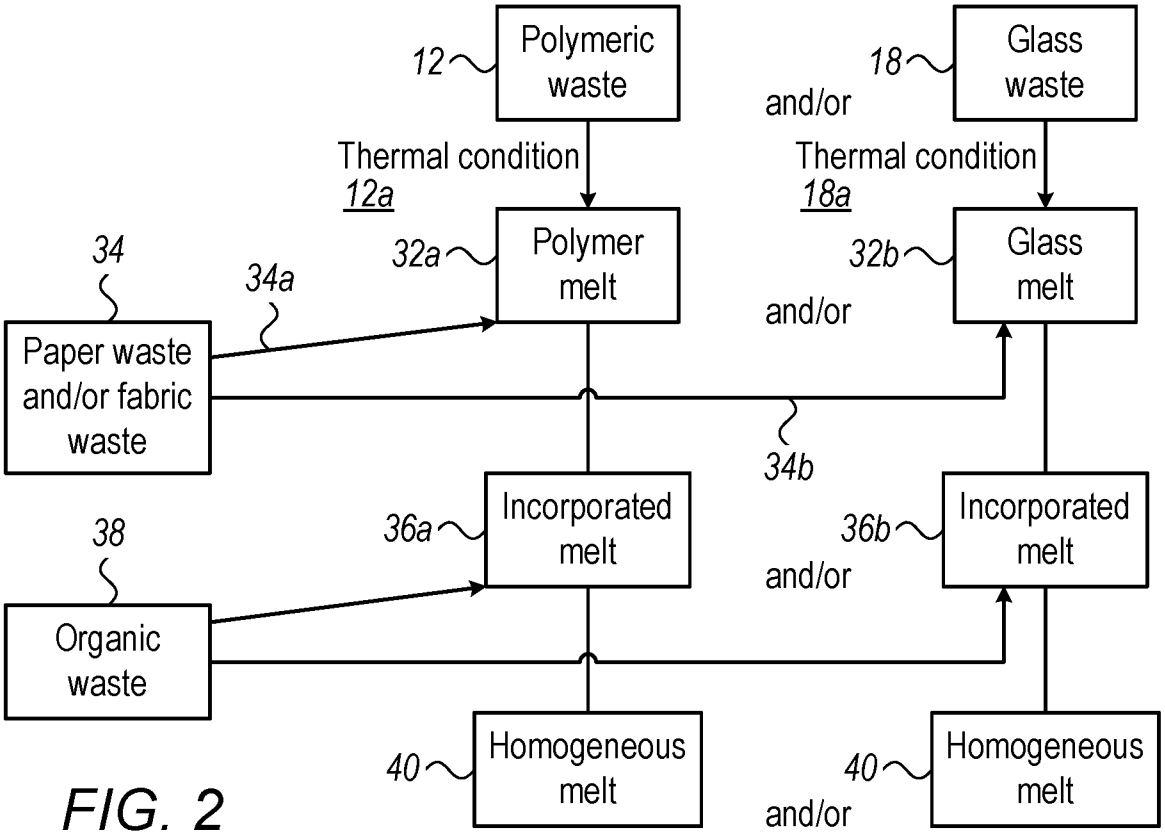

12
Polymeric waste
Thermal condition
12a
32a
Polymer melt
18
Glass waste
and/or
Thermal condition
18a
32b
Glass melt
and/or
34
34a
Paper waste and/or fabric waste
34b
36a
Incorporated melt
36b
Incorporated melt
and/or
38
Organic waste
40
Homogeneous melt
40
Homogeneous melt
and/or

FIG. 2

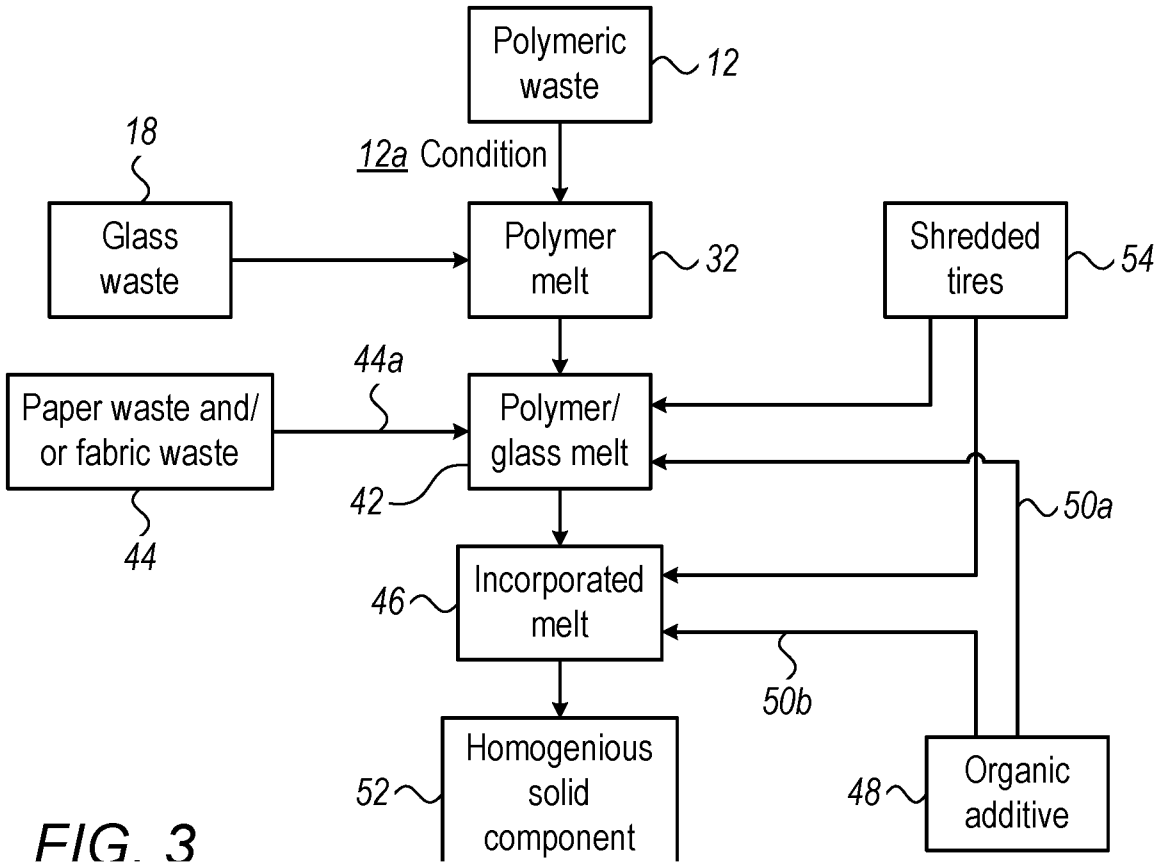

Polymeric waste
12
18
12a Condition
Glass waste
Polymer melt
32
Shredded tires
54
44a
Paper waste and/ or fabric waste
Polymer/ glass melt
42
50a
44
46
Incorporated melt
50b
Homogenious solid component
52
48
Organic additive

FIG. 3

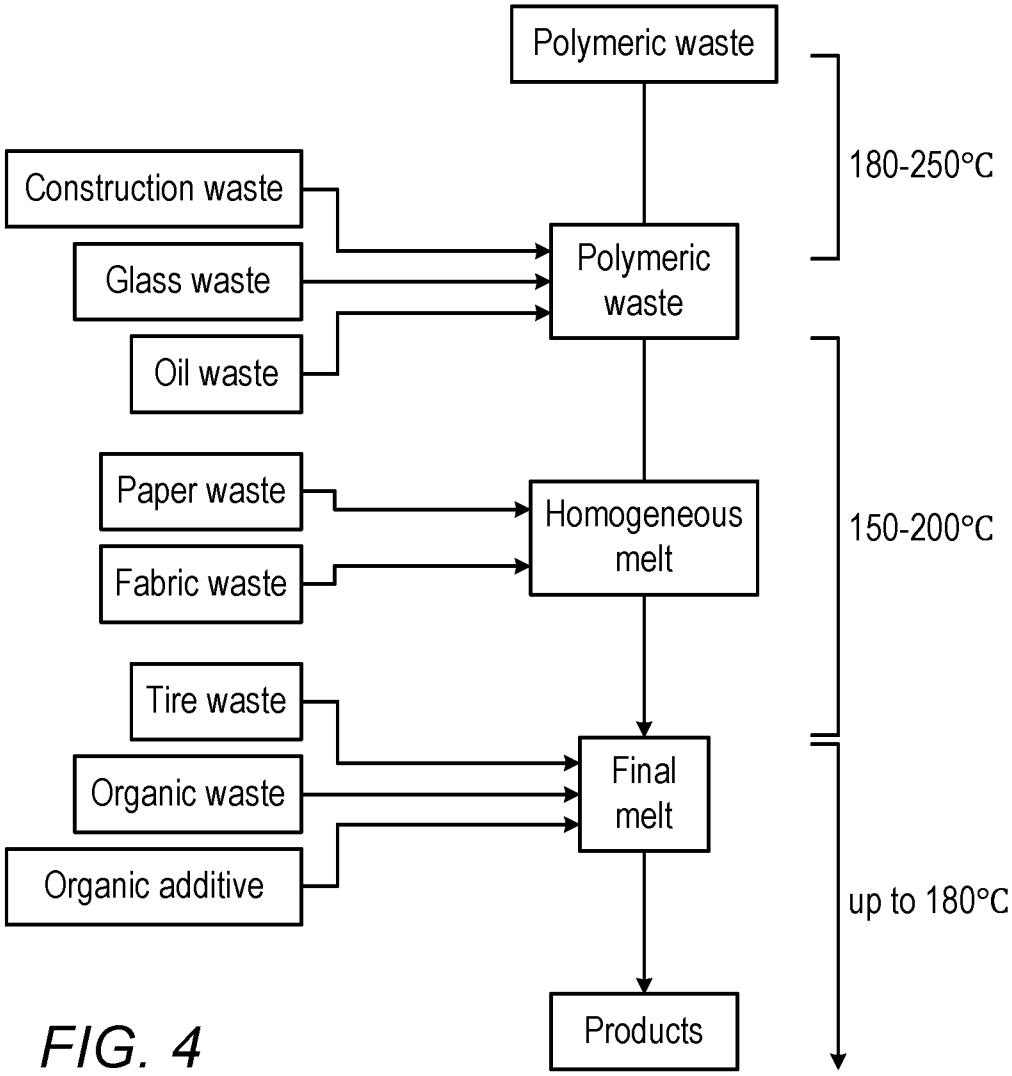

Polymeric waste
180-250°C
Construction waste
Glass waste
Oil waste
Polymeric waste
Paper waste
Fabric waste
Homogeneous melt
150-200°C
Tire waste
Organic waste
Organic additive
Final melt
up to 180°C
Products

FIG. 4

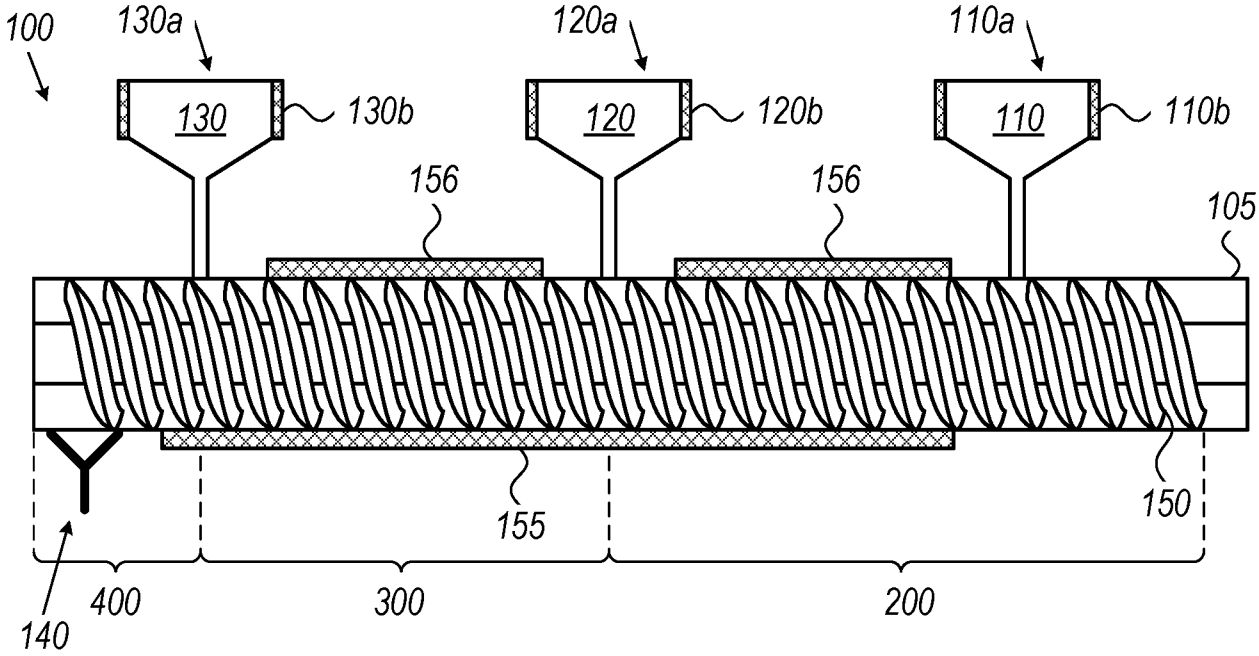

100
130a
130
130b
120a
120
120b
110a
110
110b
156
156
105
155
150
140
400
300
200

FIG. 5

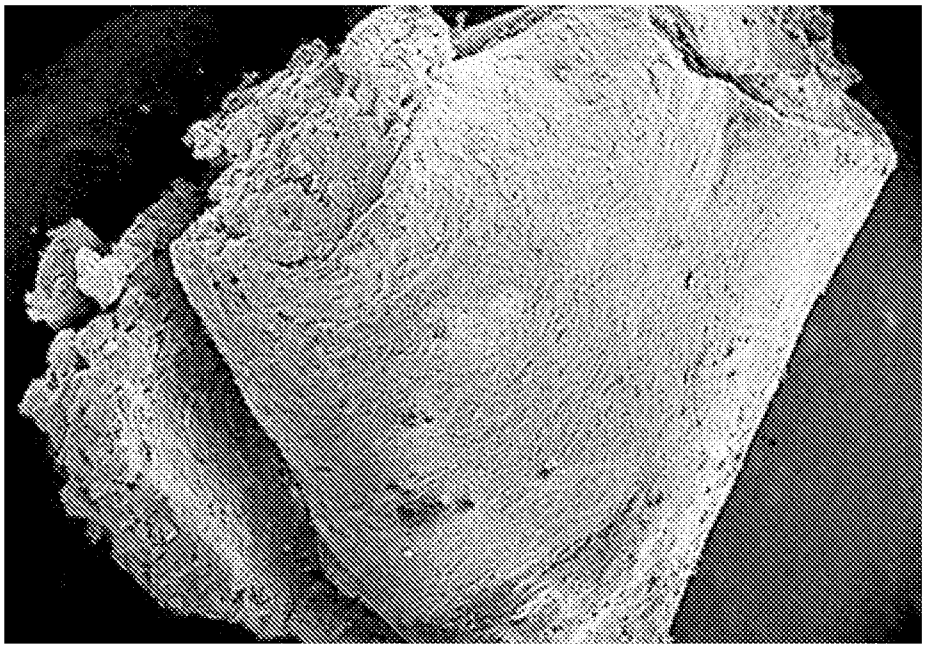


FIG. 6A

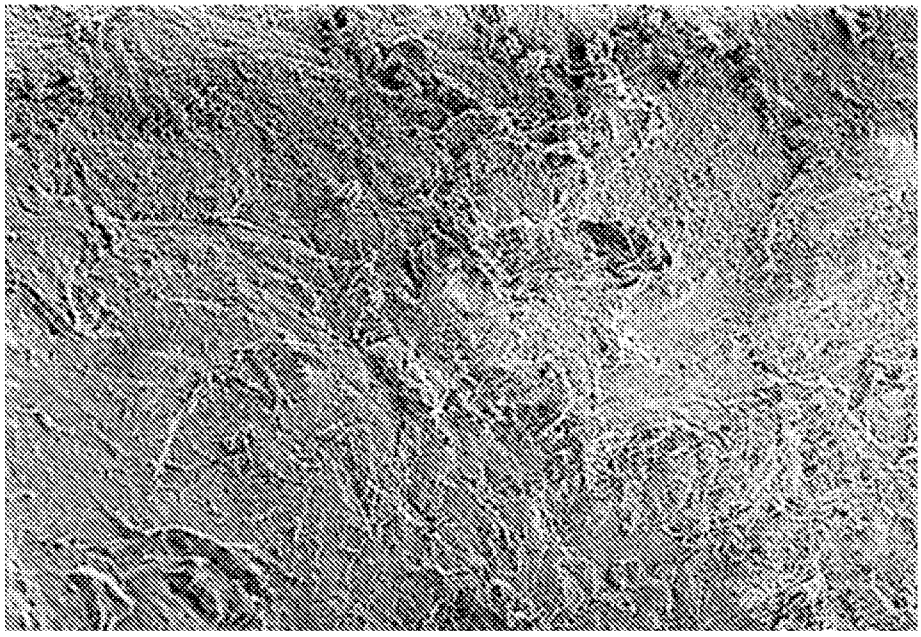


FIG. 6B

INTERNATIONAL SEARCH REPORT

International application No
PCT/IL2023/050199

A. CLASSIFICATION OF SUBJECT MATTER
INV. B29B17/04 B29B17/00 B29B7/82 B29B7/42 B29B7/88
B29B7/60 B29C48/285 B29C48/29 B29C48/793 B29C48/875
B29C48/395 B29C48/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
B29B B29C B29K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2008/128933 A1 (PRZYBYLINSKI JAMES P [US] ET AL) 5 June 2008 (2008-06-05)	69
A	figure 1 and the corresponding part of the description, asa well as paragraphs [0040] to [0050] and table B-1.	1,3-54
A	US 2003/065082 A1 (BLANKEN THOMAS CAREL [NL] ET AL) 3 April 2003 (2003-04-03) the whole document	1
X	DE 32 33 416 A1 (KALWAR KLAUS [DE]) 3 May 1984 (1984-05-03)	58-68
Y	figure 3 and the corresponding part of the description	58-68

-/--

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
5 May 2023	07/07/2023

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Rüdiger, Patrick

1

Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IL2023/050199

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:

1, 58-68(completely); 3-54, 69(partially)

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No
PCT/IL2023/050199

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 5 401 154 A (SARGENT MICHAEL M [US]) 28 March 1995 (1995-03-28) figure 1 and the corresponding part of the description	58-68
X	JP 2019 181825 A (ASAHI KASEI CORP) 24 October 2019 (2019-10-24) figure 1 and the corresponding part of the description	58-68
Y	US 2010/327728 A1 (SAITO SHINTARO [JP] ET AL) 30 December 2010 (2010-12-30) figure 1 and the corresponding part of the description	58-68
Y	DE 10 2004 040185 A1 (LEISTRITZ EXTRUSIONSTECHNIK GM [DE]) 2 March 2006 (2006-03-02) figure 4 and the corresponding part of the description as well as paragraph [0002]	58-68
Y	US 2016/271836 A1 (ORME CHESNEY RICHARD [GB]) 22 September 2016 (2016-09-22) figure 1 and the corresponding part of the description	58-68
Y	US 2021/387380 A1 (BOCK STEFAN [CH]) 16 December 2021 (2021-12-16) the whole document	58-68

1

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/IL2023/050199

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2008128933 A1	05-06-2008	CA 2670351 A1	05-06-2008
		CN 101657308 A	24-02-2010
		EP 2114648 A1	11-11-2009
		US 2008128933 A1	05-06-2008
		WO 2008066751 A1	05-06-2008
US 2003065082 A1	03-04-2003	AT 307097 T	15-11-2005
		CN 1527804 A	08-09-2004
		DE 60206749 T2	29-06-2006
		DK 1395527 T3	07-11-2005
		EP 1395527 A1	10-03-2004
		ES 2247349 T3	01-03-2006
		MX PA03010149 A	10-03-2004
		US 2003065082 A1	03-04-2003
		WO 02090288 A1	14-11-2002
DE 3233416 A1	03-05-1984	NONE	
US 5401154 A	28-03-1995	NONE	
JP 2019181825 A	24-10-2019	JP 7026559 B2	28-02-2022
		JP 2019181825 A	24-10-2019
US 2010327728 A1	30-12-2010	CN 101992551 A	30-03-2011
		JP 2011026579 A	10-02-2011
		KR 20110001920 A	06-01-2011
		TW 201119824 A	16-06-2011
		US 2010327728 A1	30-12-2010
DE 102004040185 A1	02-03-2006	NONE	
US 2016271836 A1	22-09-2016	EP 2917015 A1	16-09-2015
		US 2016271836 A1	22-09-2016
		WO 2014072674 A1	15-05-2014
US 2021387380 A1	16-12-2021	CA 3115721 A1	30-04-2020
		CN 112912219 A	04-06-2021
		DE 112019005331 A5	09-09-2021
		EP 3870418 A1	01-09-2021
		US 2021387380 A1	16-12-2021
		WO 2020083421 A1	30-04-2020

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1, 58-68(completely); 3-54, 69(partially)

   Process, in which each of the first, second and further streams is provided at a different temperature.

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2. claims: 2(completely); 3-54, 69-72(partially)

   Process in which at least one of the waste materials used is an organic-based waste material.

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3. claims: 55-57(completely); 38-54, 69-72(partially)

   Process in which yeast, a carbohydrate and/or a natural gum is used as the organic additive.

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4. claims: 51-54, 70-72(all partially)

   Product characterized by being in aggregate form.
